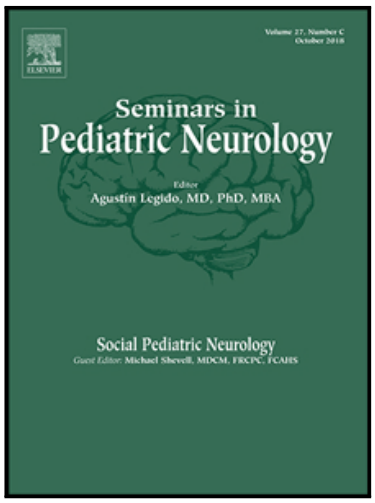
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Cerebral Visual Impairment and Clinical Assessment: The European Perspective

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Abstract

This paper summarizes the multidisciplinary pediatric assessment methods of three European centers for identifying and assessing CVI in childhood. It describes a comprehensive neurodevelopmental assessment evaluation in which visual aspects play an important part is crucial. Developmental trajectories and the heterogeneity of the clinical picture are emphasized. Multidisciplinary ophthalmology and neurodisability/neurology teamwork together with the parent and teachers to reach an integrated and individualized perspective for the individual child. This comprehensive assessment or evaluation is the starting point for habilitation programs and intervention that can support and meet the child's needs and help them reach their optimal potential. Future developments in classification of the CVI conditions, building on the child's individual assessment profile, will further enhance the direction of clinical, educational and research progress.

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Introduction

Over the last decades, Cerebral Visual Impairment (CVI) has become a major issue in children with neurodisabilities.[1] Since its description in the 1980's, the substantive definition of CVI has not undergone many changes but there remain many controversies.[2, 3] Ongoing issues have been raised and are yet to be resolved. First of all, the definitions available are variable and broad and may encompass many subtypes of basic and higher vision including differing degrees of acuity loss and visual perception deficits. On the other hand, the definitions available are not always clear enough so that conditions affecting the eye itself might or might not be included. Lastly, there is a difference in terminology between the US and the European countries with a favor in the latter for the term "Cerebral". As a significant proportion of children presenting with CVI have periventricular white matter disease, European Centers prefer the term "cerebral" to take into account cortical, sub-cortical, and white matter structures.[4, 5] A recent systematic review investigating the different definitions of CVI found fifty one papers that met inclusion criteria. The most commonly appearing themes were 'visual impairment', 'retro-chiasmatic pathway damage' and 'normal/near normal eye health'. The most consensual definitions identified in this review may not be the most empirically useful definition for describing childhood CVI and the authors proposed that a way forward was to use the definition of CVI as 'a verifiable visual dysfunction which cannot be attributed to disorders of the anterior visual pathways or any potentially co-occurring ocular impairment'.' This would be sufficiently broad and inclusive of primary cerebral or posterior visual pathway origins of visual dysfunction, whilst also requiring a systematic and explicit means of 'verifying' the disorder through appropriate clinical assessment methods.[2]

Whereas the definition of CVI is broad, the clinical spectrum is even broader and so is the array of underlying conditions causing CVI, which can be acquired or congenital, occurring in the pre-, peri-, or postnatal period.[6–8] Many conditions have been associated with CVI, such as cerebral malformations, hydrocephalus, central nervous system infections, epilepsy, head trauma, metabolic and neurodegenerative diseases, or intoxication with drugs or poisons.[9]

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However, cryptogenic cases have also been described.[10] Still, the most frequent conditions linked with CVI are periventricular white matter disease, typically occurring in preterm infants, and hypoxic-ischemic encephalopathy around birth, both of which can lead to cerebral palsy (CP). In children with CP, up to 70 % have basic and higher visual dysfunction including visual perceptual problems.[11,12] The first publications on visual dysfunction in CP described the children as having object recognition difficulties.[13,14] However, recent studies have shown that children with CP also suffer from visuospatial deficits impacting on their locomotion.[15]

As severe visual impairment of any etiology has an important impact on other developmental domains including cognition, early diagnosis is of utmost importance to prevent cumulative cognitive, motor, socio-emotional, and educational deficits.[16] The greatest delays are found in infants and young children with profound vision loss and therefore early infancy vision promotion is recommended for those with early severe vision reduction.[17]

An ophthalmological, neuropediatric, developmental, vision and visuomotor assessment is required, in addition to interviewing the parent and other professionals involved in the child's medical and educational care. This assessment will underpin the identification and diagnosis of whether the child has CVI or not, identify the child's strengths and needs and provide important information for the child's care, habilitation and education plan. As CVI has so many aspects, habilitation will have to be multidimensional and individually tailored.

An optimal habilitation plan thus needs a thorough systematic, multidimensional, and multidisciplinary assessment. The importance of a systematic means of screening for and assessing CVI has been emphasized in the recent systematic review.[2] A number of different European pediatric centers, including London (UK), Brescia (Italy), and Leuven (Belgium), have been developing multidisciplinary methods of assessment in the clinic and research context drawing on ophthalmology, neurology and neurodisability models of care. In the following sections of the paper we consider our assessment models in the context of clinical

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and research knowledge of the developmental trajectories of CVI, the heterogeneity of the clinical picture, assessment methods, co-morbidities and classification of CVI and in the light of defining an optimal habilitation plan.

Thoughts to consider in the approach of CVI

Need for early identification

Early identification of CVI is critical because it can lead firstly to early detection of infants at risk of brain damage and, secondly, to implement intervention in order to gauge plasticity and recovery from perinatal brain damage early in life through measured changes over time.[18]

Developmental trajectories and outcomes of CVI

An important issue to consider in the screening and/or assessment of CVI in children is the developing brain and related developmental trajectories in typically developing children. Along with the maturation of lower visual functions (with visual acuity reaching an adult level between three and five years of age), various higher level visual functions complete their development at differing and specific ages. For example, already from the age of six weeks, recognition of faces is based solely on the internal configuration of the face.[19] At 3 months of age, infants show preferential attention for upright human movement, which is crucial for the ability to recognize people and to make social contacts.[20] Meanwhile, within the first 16 months, form- and motion-processing abilities start to develop but will not be completed before the age of 4 years.[21] Many higher vision abilities continue to develop throughout childhood, such as complex form-ground discrimination and visual attention, which continues to mature until school age.[22] These higher vision abilities develop alongside other cognitive abilities, with higher visual perceptual abilities developing in accordance with brain maturation and developmental age.[23]

So, in contrast to adults where a lesion in the brain almost certainly leads to permanent deficits, in children there is hope for improving outcomes and even recovery in some cases

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as injured or compensatory neural networks develop over time.[24, 25] For example, whereas in the acute phase following a traumatic event, children often seem to be cortically blind, functional improvement/recovery has been reported after variable timeperiods.[26]

However, in most patients with CVI normal visual function will not be regained, as the vast majority of them have associated neurological and ophthalmological deficits, including intellectual disabilities, CP, seizures, microcephaly, hydrocephalus or progressive central nervous system (CNS) degeneration.[3,10] In addition, CVI is often associated with ophthalmologic abnormalities like strabismus, refraction disorders, nystagmus, and the presence of large optic disc cupping often associated with generalized optic disc pallor.[27] Improvement of vision nevertheless can be seen and in some cases of CVI the degree of reduction in visual acuity is quite mild.[10, 28] However, many clinicians and researchers state that children with CVI and additional neurologic damage have the least favorable prognosis for normal development of basic (and higher) vision abilities.[9]

Value of neuroimaging

Areas of brain damage may be focal or widespread and variable and the MRI brain scan is not of diagnostic value in itself. A correlation between neurovisual and neuroradiological findings is not always present. In fact, a normal MRI does not exclude the presence of CVI.[5] Many studies have reported an association between visuo-spatial dysfunction and occipito-parietal periventricular white matter lesions but no significant association between visuo-perceptual dysfunction and occipito-temporal white matter damage was found.[12, 29, 30] There is some agreement among investigators that the lack of correlation between higher visual skills and observed morphological brain anomalies may be explained by the fact that conventional MRI techniques do not reveal macro-structural injuries within the visual pathways.[5, 30] In this direction, the development of new neuroimaging advanced techniques such as diffusion tensor imaging (DTI) has proven to be highly useful not only as a method to

characterize white matter macro-structural damage, but also to identify potential biomarkers for visual processing impairment.[31–33]

Heterogeneity of the clinical picture

Children with CVI present a variable symptom complex. For example, children can exhibit severe CVI impairment with complete loss of visual awareness, whereas children with milder CVI may only exhibit visual perceptual problems like the inability to recognize faces or to visually identify objects.[22] This depends on the timing, location and extent of the causative lesion itself. Ortibus et al. have shown that more extensive white matter damage is associated with more important object recognition difficulties, coined by Dutton as 'dorsal-stream dysfunction plus'.[5, 34] In more recent work, several groups have shown that global lesion extent and specific thalamus involvement, measured by a semi-quantitative scale, define the clinical picture. [35, 36]

Traditionally, clinical symptoms of CVI are categorized according to the proposed underlying neuroanatomical distribution of the visual functions as first described in adults.[37] As such, some symptoms could be related to primary visual pathways, some symptoms have been more related to putative ventral stream deficits and other to dysfunctions of the dorsal stream. For an overview of clinical symptoms, we refer to excellent overviews such as those of Dutton et al. and Fazzi.[30, 38] Typically not all symptoms occur at the same time and they can evolve over time.[9,39] This could be due to recovery and plasticity mechanisms. In addition, clinical experience teaches that the child may also acquire compensation strategies for the visual deficits. A classic example is the dependence on color for recognizing objects or finding pictures on communication charts by remembering their location. Lastly, as described in congenitally blind people, the other senses may become more superior and non-visual perceptual processing may actively recruit the occipital areas.[40]

Multidisciplinary approach for CVI: the European model

A first crucial step in early identification of CVI is consideration of the child's pediatric medical history, including prenatal, neonatal and early post-natal history, and later traumatic episodes, which will alert one to the risk of congenital or later CVI. The majority of tools used to identify CVI were not originally developed to diagnose this condition. Both clinically and in research, groups of assessments are often used together to address different parts of the diagnostic process, which will draw on different disciplines including pediatric ophthalmology, optometry and orthoptics, developmental pediatrics, neuroimaging, visual electrophysiology, clinical neuropsychology, and occupational therapy. These may also be supplemented by education. Individually, such assessments are unlikely to reach sufficient specificity or sensitivity to accurately differentiate between CVI and other conditions. However, when taken together and combined into a final integrated formulation, the multidisciplinary team and disciplines using a combination of assessment methods may successfully identify CVI in the childhood population at different ages and levels of vision and cognitive ability.[41] This combined approach is however only found in a few pediatric centers, such as the Developmental Vision Clinic of Great Ormond Street Hospital, London (UK), the CVI clinic in Leuven (Belgium) and the Center of Child Neurophthalmology at the Unit of Child Neurology and Psychiatry, ASST Spedali Civili of Brescia (Italy), where a multidisciplinary pediatric neurodisability/developmental vision clinic team works alongside pediatric ophthalmology.

Screening questionnaires

Several questionnaires have been developed for screening purposes, to obtain an initial idea of whether children are at risk for CVI, depending on their clinical characteristics.[42–46] Some of them have been developed for specific age ranges, others for specific risk groups such as children with CP or for a specific geographical population. Although their clinical screening value is recognized, not many of these questionnaires have been studied for reliability or validity. Studies of the Preverbal Visual Assessment (PreVias) for example show the capacity of the questionnaire to differentiate typically developing children from those with slower

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developing visual abilities.[47] However, it is not possible to specifically identify the children with CVI from other conditions associated with slower maturing of vision abilities, for example as in prematurity. Ortibus et al. have validated their questionnaire in a sample of children seen at a CVI clinic and tested with an object recognition battery. Although relatively good sensitivity and specificity of object recognition could be found based on 7 specific items, this set of questions could not reliably give information on whether a child is also at risk for visuo-motor or visuo-spatial difficulties.[42] Moreover, some of the items are not suitable for children with a motor disability, which is why Salavati and colleagues developed a questionnaire tailored to this population. Validity studies were performed, however, the criteria for CVI diagnosis in their population were not so clear.[43] Philip et al. meanwhile rephrased some of the questions of an inventory developed by the group of Dutton, so as to fit the specific conditions of India.[46] Taken together, no one questionnaire seems reliable and valid enough at the moment to initially detect or screen children at risk for CVI and a combination of questionnaires and/or addition of information from parents and other caregivers and pediatric clinicians involved with the child is likely to be necessary. Further research and development of validated and worldwide useful questionnaires is urgently needed. Screening identification will then need to be followed by the comprehensive multidisciplinary assessment.

Cognitive/developmental assessment

Before progressing to assess higher visual perceptual abilities, it is necessary to evaluate the child's general cognitive and adaptive behavioral levels using standard assessment tools. If the child has very low visual acuity then it is advisable to use the only available semi-standardized scales for children with visual impairment (the Reynell Zinkin Scales for young children with visual handicap) for under 5s or the verbal subscales of cognitive schedules for school aged children so that the cognitive assessment is not compromised by vision reduction.[48] Children with better vision can be assessed on infancy and childhood developmental assessment scales that include verbal and non-verbal scales. This will

establish whether the child has significant developmental delay or intellectual disability by the time the child is at preschool or school age.

Children with complex disability who are functioning in the severe-profound intellectual disability range are unlikely to be able to participate in standardized visual perceptual assessments, which require a developmental age of at least two to five years, depending on the test. If the child has the cognitive ability to undertake a visual perceptual assessment, it is anticipated that the visual perceptual abilities are in line with their level of cognitive development, so any significant discrepancy and reduction in vision perception can be assumed to be a selective visual perceptual deficit.

Assessment of vision deficits

Children with CVI present with specific difficulties in visual behavior or function. Although they have a normal or quite normal eye examination, they show poor visual reactions and poor visual attention. The presence of visual latency, a limited visual attention span and extremely shortened gaze behavior often result in misjudgment of these children as inattentive, incapable or poorly motivated.[49, 50] Other important features of CVI are oculomotor disorders such as nystagmus, unstable fixation, discontinuous smooth pursuit, and/or abnormal saccades. Also, they may show reduced visual acuity, poor depth perception and visual field restriction.[51] This could explain their characteristic head turn when looking at or reaching for an object. In addition, they tend to use peripheral vision to look at an object and appear not to use their central vision.[38] Also, children with CVI often lean forward to bring their eyes close to an object to reduce the 'crowding' effect or to improve resolution. This way, the amount of 'noise' in the background is reduced. However, this phenomenon is often misinterpreted as task rejection or as rebellious misbehavior.[52] Paradoxically, in nearly one-third of the children photophobia (intolerance of light) is present, however gazing at lights is very characteristic, particularly in those with additional complex disability.[6] Although perception of movement and color are frequently preserved in children

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with CVI, they show a preference for brightly colored objects (specifically red and yellow).[28] Typically, they are better off with higher contrasts and they do have problems with perception of motion coherence as shown in many studies in preterm infants.[21, 53] Vision for figure-ground discrimination and crowding or simultanagnosia appears to be particularly compromised, whatever the child's level of visual acuity.

Basic functional vision

The newborn infant has a measurable visual acuity and contrast sensitivity just hours after birth and uses vision to explore and learn about the environment.[54] Measurements of visual competences, becoming more efficient during the first weeks of life, can help to identify infants at risk of CVI. In those at risk, such as very preterm infants, early signs of CVI that can be identified are abnormal visual behavior, low visual acuity and visual field, and malfunctioning of basic visual functions, particularly oculomotor functions.[9]

The feasibility of testing early visual function has increased considerably. Ophthalmologists/optometrists or neurodisability pediatricians can now use several tools to assess basic functional vision in newborns and infancy and those with or without early signs of other disabilities, particularly CP and cognitive disability.[55–58] Most recently, the group from Fazzi at Brescia, showed that early detection of visual function deficits could be a means of rapidly assessing visual and neurological status in infants born preterm.[58]

From the early years and through the childhood years, standard measures of visual detection and acuity, contrast sensitivity, color vision and stereopsis should be undertaken in the clinic environment as well as direct assessment of oculomotor behavior, saccades and visual field perimetry. These assessments will establish the child's level of vision and the degree of visual impairment that may be hindering interaction with the environment or impact on depth perception and movement in space. In addition, the results are important in order to reliably use neuropsychological measures. For example, a good acuity is necessary to be able to

recognize objects on a PC screen or to undertake a puzzle or visual-perceptual test. When contrast sensitivity is reduced, materials should be presented in an optimal way.

The vision work-up thus includes measurement of resolution visual acuity such as preferential looking grating acuity (e.g. Teller acuity cards) or logMAR acuity (e.g. Sonksen logMAR tests).[59, 60] For very young children from birth or those with very low levels of vision or complex multidisability, the Near Detection Scale is a useful standard measure of basic detection from no vision to detection vision of 'lures' of diminishing size at standard distance.[61] Visual field perimetry can be a challenge to test in children with a complex neurodisability. Behavioral field testing is more likely to be reliable than standard perimetry testing in those children.[62] Oculomotor assessment should include the detection of strabismus and further evaluate fixation (absent, sporadic, unstable, stable), smooth pursuit (absent, discontinuous, normal) and saccadic movements (abnormal, normal).[50] Refraction errors should be sought for as early as possible. Meanwhile, the development of playful stereopsis testing in young children has been launched: the Accurate STEReoacuity measurement On a mobIle Device (Asteroid) allows for stereoacuity testing through a game for health.[63] For school aged children, the Frisby stereo test is suitable.[64] As visual attention is a fundamental primordial requisite for and association of good overall visual function, its assessment is very important. In children under the age of 4 years, observational methods are used, due to the lack of standardized instruments which have been developed for older children.[65, 66]

Neuropsychological assessment of visual perception

The value of a standardized neuropsychological assessment of visual perception abilities is that it permits comparison with the test population norms (in the test manual), or reported typically developing community populations. This leads to identifying whether the child is in line with or behind normative (developmental) age expectations. Also, longitudinal follow up

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can demonstrate changes in the individual child's trajectories. Based on available tests, evaluation can be successfully performed in children from ages of 2-5 years to adulthood.

Specific instruments have been developed for specific visual functions, such as the line bisection task to detect hemi-neglect or the embedded figure test to specifically test figure-ground detection. These tests however primarily target recognition of abstract materials. Instruments such as the Test of Visual Perception (TVPS-3), the Developmental Test of Visual Perception third edition (DTVP-III), the NEPSY (Developmental NEuroPSYchological Assessment) and the L-Post (the Leuven Perceptual Organization Test) consist of several subtests and scales that tap into a range of different visual functions.[67–70] The latter is an online test, in which 15 subtests cover a range of processes, such as figure-ground segmentation, local and global processing, and shape perception. To reduce cognitive load, a matching-to-sample task was designed.[71]

Many tests however are not reliable in children under the age of 5 years or in children with complex disability. This can be explained by limitations of the test (inappropriate for age) or because the child presents with limitations such as reduction in visual acuity, speech or cognitive delay or motor impairment. The lack of adequate testing tools in children with complex neurodisability makes the diagnosis of visual and intellectual disability in these children much more reliant on skilled clinical judgment.

If existing, assessment batteries of visual perception for the child under 5 years have not been widely distributed. One example is the L94, an object recognition battery that is used throughout Flanders in Belgium.[72] Because of the lack of instruments tailored to even younger children, the group of Ortibus et al. started developing assessment batteries for children aged 24 and 30 months.[73] The test focuses on object recognition in black and white

and colored forms, in 2D as well as in 3D and in crowded and uncrowded situations. Similarly, because of the L94 becoming outdated and the fact that it is purely based on recognition of black and white line drawings, the same group embarked on the development of a screening instrument for CVI in children between 3 and 6 years old, following a matching to sample paradigm along the lines of the L-Post. In the same line, this instrument also taps into different visual perceptual processes. Important in the development of these new instruments is to carefully study their reliability, including test-retest reliability, and their discriminative validity. The question pursued in the authors' research is whether they are able to discriminate children with CVI from those with deficits on the tests due to other conditions such as a pure ocular condition or intellectual disability.[74]

Certain subtests are found to be particularly discerning of visual perceptual dysfunction in children who are preterm, such as figure-ground, spatial relations, closure and tests of discrimination.[75] There is high interest in figure ground discrimination which is considered a dorsal stream deficit and object recognition and stereopsis which is considered a ventral stream deficit.[76, 77] However, clinical and research experience shows that children with potential CVI and pediatric neurodevelopmental risks may have very uneven profiles on assessment batteries including visual perceptual ones.[5, 50, 78] It is therefore important to remember that no single subtest or reduced scale is likely to be sufficient to cover the range of strengths and deficits that may be part of the individual child's profile.

Visuomotor examination

As vision has a profound impact on motor development, assessment of visuomotor abilities or deficits must not be overlooked.[79, 80] Young children for example could be evaluated by means of the Peabody Developmental Motor Scales (PDMS-2), which are available from birth through 5 years of age and cover reflexes, stationary motor abilities, locomotion, object manipulation, grasping and visual motor integration and measuring interrelated motor abilities.[81] The Movement Assessment Battery for Children (Movement ABC-2) is now

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available for 3 year olds upwards and has had decades of clinical and research use and has reliable norms. It covers areas such as manual dexterity and aiming and catching.[82] Other tests like the Beery-Buktenica Developmental Test of Visual-Motor Integration Sixth Edition (Beery-VMI) are available from 2 years upwards and this is supplemented by a visual perception and motor coordination test which are designed to compare the child's test results with relatively pure visual and motor performances.[83] Important aspects such as reaching for objects to rule out optic ataxia or limb apraxia can be evaluated for example by the letter box test of L Hyvarinen or the posting coins subtest of the Movement ABC-2.[82, 84, 85] Braddick and colleagues have also reported on the factors that play a role in the development of reaching. For example, as children grow older, but also depending on their reaching experience, they use smoother trajectories to reach and develop a preference for smaller over larger objects.[79] In children with CP and other children with CVI, spatial navigation can be a real problem, prohibiting normal navigation despite relatively good motor abilities.[86] Real world experience, such as walking along a corridor in the hospital center or along the street, may be important for getting an idea of the orientation and mobility abilities of the child.

The role of environment

Critical for a good appraisal of the visual stimuli for children with CVI is a controlled, clutter free environment, good lighting and optimal contrast.[87] However it might be revealing to also assess or observe the child's visual function in less optimal conditions such as a busy classroom or a noisy playground. When tired, the child's performance might be compromised and for those with a motor disability, optimal seating positions are required, as to insure trunk stability and consequently, better hand performance. It has to be verified first that the child's peripheral visual abilities are satisfactory or need to be compensated for, e.g. when having a hemianopia.

Visual function and functional vision

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Assessing visual function describes how the child's eyes and his or her basic and higher visual systems function. It sheds light on the ability to process visual stimuli of a particular dimension, e.g. form, colour or motion.[1] Functional vision on the other hand describes how a person functions with his or her visual abilities and, more importantly, looks at how vision determines function in everyday life.[88] As such, it is important to consider the impact of the child's vision on communication and social relations, orientation and mobility, daily life activities and for sustained near vision tasks. Communication in daily life not only happens between person to person but also in busy environments or from a greater distance. For instance, difficulties with crowding will particularly impact the child's ability to find their parent or caregiver or friends in a busy environment, affecting their social relating at school. The final consideration is how to assess the needs of a child with a multiple disability, in whom standardized assessment may not be feasible. In this group, we rely considerably on history-taking and information gathering from parents and educational and health professionals who have had a sustained relationship with the child. In these cases, a detailed observation of the child's vision, general play and social interactions in the clinic or everyday environment, is crucial, using standardised observational frameworks where possible. Observation in optimal and less optimal conditions and videotaping, will enable giving detailed feedback to parents, focussing on the family's and educator's concerns.

Comorbidities in children with CVI

As most of the children with CVI have a brain lesion, comorbidities are markedly present. In addition to the well-known association between CP and CVI, many children present with uneven cognitive or learning profiles, intellectual disability, attentional difficulties/ ADHD and also autistic spectrum disorder (ASD) and characteristics of developmental coordination disorder (DCD).[87] There is ongoing research interest in whether this raised tendency is a reflection of common pathways at risk such as 'dorsal stream deficits'.[89] For example, coherence problems and difficulties in local/global processing (in ASD) and clumsiness (in DCD) are also apparent in children with CVI.[89, 90] Children with DCD have motor imagery

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problems and problems with estimating distances, leading to clumsy reaching and wrong pre-position of the hand, which is also observed in children with CVI.[91] As such, while assessing a child with suspicion of CVI, a multidisciplinary, multidimensional and integrated assessment approach is necessary to identify, differentiate and consider the impact of other potential comorbidities. On the other hand, there is also a need for understanding of the impact of CVI on the global functioning of the child and certain features, for instance associated with DCD or ASD, may in some cases arise from this. For example, clumsiness could also be related to impaired visual function and children who show anxiousness or lack of eye contact are not necessarily children with ASD.[92, 93] In summary, in children suspected of CVI, all potential developmental problems and needs should be considered as part of the differential assessment of the total profile of the child for the most beneficial and realistic support and intervention. This requires an evolutionary assessment process at times as developmental features may change as the child matures and it becomes more clear whether there is an additional comorbidity or not. This will also assist the parent in understanding more how CVI impacts on the child and the effects of other comorbidities such as intellectual disability or ADHD.

Classification of CVI

Increasing attention is now being paid towards classification and sub-typing of CVI. This is in response to the growing realization that the heterogeneity of CVI requires more individual sub-classification in order to identify needs better, establish greater consensus in the field, and reach more informed knowledge through clinical assessment and research.[2] As shown in a study by Ortibus et al, depending on the extent of the lesion, particularly when periventricular white matter is present, children present with either dorsal stream symptoms only or a combination of ventral and dorsal stream deficits.[5] Dutton even coined this 'dorsal stream dysfunction plus'.[34] Indeed, Atkinson et al. have shown that the maturation of the dorsal stream runs a protracted course compared to the ventral one, making it more vulnerable to damage.[94] Several authors have proposed different classification systems for

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CVI, including for instance the basis of variable academic and cognitive abilities of the children.[87] However, even within these groups, separate subgroups can be distinguished. In a study by Sakki et al. from Dales' group in London, classification analyses including cluster analysis was undertaken of the multidisciplinary standard clinical assessment measures of basic and higher vision abilities in children with diagnosed or suspected CVI (aged 5-16 years). The data driven groups highlighted a gradient of severity of CVI, with some similarity to Philip and Dutton's proposed classification.[95] Ophthalmological, basic and higher vision, and cognition were all in a gradient of severity and breadth of disorders, with more selective deficits conventionally associated with dorsal stream vulnerability in the highest functioning group (Sakki et al. 2018, personal communication).

Advances in classification should thus drive both clinical developments and research knowledge and understanding of CVI in the field. It may also show the best multidisciplinary assessment tools and battery for identifying CVI and the different groups in the clinic environment. However, for the individual child, the continuing need will be for the comprehensive assessment approach described in this paper to ensure that the individual child's profile of strengths and needs is identified. Good communication and partnership approaches with the parent, teachers and therapists will lead to ensuring that the child's identification of needs draws on the knowledge and expertise of all involved with the child and can also be implemented in the school environment.

Visual assessment: perspective towards visual and habilitative intervention

The earlier identification is likely to be in those infants and young children with greater vision reduction and this is more commonly associated with more pervasive brain injury and complex multi-disability. This has implications for early intervention as the milder end of the CVI spectrum is often not identified until the child is late preschool or school age. Each child will need their assessment to be combined with individual guidance and habilitation approaches and strategies to support the child's development, function, and adaptation in

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their everyday life situations, including home and school. Further assessments can be undertaken in the future to update the record of the child's needs and to establish the success of support and interventions and also to provide guidance on further requirements for the child.

As the picture of CVI is so heterogeneous, no single type of treatment is applicable to all children with CVI, nor is any individual treatment successful with all. In addition, there are few scientifically robust contemporary theories or scientific knowledge to guide visual intervention with this population. After the first publications of Nathalie Barraga in the 1960s, vision rehabilitation became common practice in teaching and in early intervention programs for children with visual impairments. These rehabilitation programs replaced the prevailing philosophy of ''sight saving,'' because it became apparent that vision would not be further impaired by use.[96] The only randomized controlled trial in the field of early intervention and visual impairment is that of Sonksen et al. from the Developmental Vision Service, Great Ormond Street Hospital, using a developmental approach to early vision promotion.[17] Although the majority of infants had peripheral or anterior visual disorders, the subsample with CVI showed similar faster progress in all vision functions if they were in the intervention arm. Meanwhile, there is growing evidence that early experience is most effective during the critical periods of brain development, where it can mould neural circuits and that repair from maldevelopment is much harder at a later stage of maturation.[25, 97] To date, there have been few systematic trials of early intervention with infants or young children with CVI and early identification and research in this area is urgently required.[98]

Currently, however, it is well-accepted that more information is needed on the efficacy of visual interventions to draw recommendation guidelines on visual deficit treatment.[96, 98] Studies looking at the effect of visual perceptual training have shown that therapy needs to be specific, focused, repetitive, and intense to have a superior effect.[99, 100] Also, new neuroscience research is beginning to focus on training of specific areas of vision

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dysfunction like figure-ground discrimination, with promising results.[101] Lastly, motivation plays a crucial role in achieving optimal results. Therefore, customized, adaptive rehabilitation may be needed for optimal treatment outcomes, taking into account the child's overall developmental status and changing needs and this may be enhanced by using serious games.[102, 103]

The central aim of habilitative programs addressed to young patients with CVI should be the promotion of those functions which can guarantee the acquisition of better levels of autonomy in domestic and extra-domestic environment. Training regarding the acquisition of autonomy in routines such as dressing, grooming, bedtime, and bathing can promote a better self- mastery and lead to a better level of autonomy in domestic and extra domestic daily living routines. Further assistance is needed for better integration into schoolmates' groups, which may include the speech and language therapist, occupational therapist, psychologist and teacher.

Philip & Dutton proposed a list of visuo-cognitive symptoms and related approaches to promote the organization of a suitable environment for these patients.[95] For school-aged children, environmental aspects, such as the context of school surroundings and lessons, should also be considered. With regards to academic skills, reading is a common and demanding visual task at school. Since it requires fixation of gaze and saccades, often abnormal in children with CVI, the access to the written word and to numbers in columns or from printed pages can be compromised. Limiting the amount of words per page or minimizing crowding with well-spaced text layouts is necessary to guarantee best performances. Copying and drawing single and composite geometric forms is another common task which is required at school and which can be difficult to manage for children with CVI. A suitable way to improve the children's performance is to show how to draw the forms and shapes, which have to be copied while the child is watching or to guide the child during the task, providing him/her with some hints, which can also facilitate partial

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successes. If the patient tends to lose spaces among words, a suitable solution is to teach children to draw a line at the end of each word along the baseline to “represent” the space needed before writing the next word.[95]

The parents’ attempts to solve the child’s problems aren’t usually enough to guarantee a positive outcome. In the majority of children with CVI, the specialist therapists’ help, including the specialist teacher (visual impairment) or occupational therapist, will promote the impaired skills, changing the child’s exposition to those tasks which are difficult to deal with. In subjects with motor planning deficits, task-oriented programs and motor-training-based interventions have shown stronger treatment effects than process-oriented therapy.[104] Task oriented treatments permit the patient to concentrate his/her resources to the learning of specific abilities. As regards visual perceptual and visual motor problems the management is twofold. First, an organization of a children’s suitable environment, which can promote a better adaptation of the patients to daily living skills is fundamental. In addition, the organization of a personalized habilitative training is also of particular importance because it can promote the emerging visual-perceptual and cognitive functions.

The child’s mental health will also need to be considered as she or he may be at greater risk, though little is yet known of this in children with CVI. Elevated mental health difficulties have been shown in recent systematic reviews of children with CP and prematurity, with or without neurosensory impairments.[105, 106] Other pediatric risks for CVI, such as traumatic brain injury, also have long term risks for behavior problems and disorders.[107]

Conclusion

This paper summarizes the multidisciplinary pediatric assessment methods of three European centers for identifying and assessing CVI in childhood. It highlights that a comprehensive neurodevelopmental assessment evaluation in which visual aspects play an important part is crucial. Multidisciplinary ophthalmology and neurodisability/neurology teams

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work together with the parent and teachers to reach an integrated and individualized perspective for the individual child. This comprehensive assessment or evaluation is the starting point for habilitation programs and intervention that can support and meet the child's needs and help them reach their optimal potential. Future developments in classification of the CVI conditions, building on the child's individual assessment profile, will further enhance the direction of clinical, educational and research progress.

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References

1. Salt A, Sargent J: Common visual problems in children with disability. Arch Dis Child 99:1163–1168, 2014.
2. Sakki HEA, Dale NJ, Sargent J, et al.: Is there consensus in defining childhood cerebral visual impairment? A systematic review of terminology and definitions. Br J Ophthalmol 102:424–432, 2018.
3. Whiting S, Jan JE, Wong PK, et al.: Permanent cortical visual impairment in children. Dev Med Child Neurol 27:730–739, 1985.
4. Jacobson L, Ek U, Fernell E, et al.: Visual impairment in preterm children with periventricular leukomalacia--visual, cognitive and neuropaediatric characteristics related to cerebral imaging. Dev Med Child Neurol 38:724–735, 1996.
5. Ortibus E, Lagae L, Castees I, et al.: Assessment of cerebral visual impairment with the L94 visual perceptual battery: Clinical value and correlation with MRI findings. Dev Med Child Neurol 51, 2009.
6. Dutton GN, McKillop EC, Saidkasimova S: Visual problems as a result of brain damage in children. In: Br J Ophthalmol. England; 2006. p. 932–933. .
7. Khetpal V, Donahue SP: Cortical visual impairment: etiology, associated findings, and prognosis in a tertiary care setting. J aapos 11:235–239, 2007.
8. Kozeis N: Brain visual impairment in childhood: mini review. Hippokratia 14:249–251, 2010.
9. Good W V, Jan JE, Burden SK, et al.: Recent advances in cortical visual impairment. Dev Med Child Neurol 43:56–60, 2001.
10. Huo R, Burden SK, Hoyt CS, et al.: Chronic cortical visual impairment in children: aetiology, prognosis, and associated neurological deficits. Br J Ophthalmol 83:670–675, 1999. Available at: http://ovidsp.ovid.com/ovidweb.cgi?T=JS&CSC=Y&NEWS=N&PAGE=fulltext&D=med4&AN=10340973%5Cnhttp://sfx.ucl.ac.uk/sfx_local?sid=OVID:medline&id=pmid:1034

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0973&id=doi:&issn=0007-1161&isbn=&volume=83&issue=6&spage=670&pages=670-5&date=1999&title=British+Journal+.

11. Ego A, Lidzba K, Brovedani P, et al.: Visual-perceptual impairment in children with cerebral palsy: a systematic review. Dev Med Child Neurol 57 Suppl 2:46–51, 2015.
12. Fazzi E, Bova SM, Uggetti C, et al.: Visual-perceptual impairment in children with periventricular leukomalacia. Brain Dev 26:506–512, 2004.
13. Cioni G, Fazzi B, Ipata AE, et al.: Correlation between cerebral visual impairment and magnetic resonance imaging in children with neonatal encephalopathy. Dev Med Child Neurol 38:120–132, 1996.
14. Stiers P, De Cock P, Vandenbussche E: Impaired visual perceptual performance on an object recognition task in children with cerebral visual impairment. Neuropediatrics 29:80–88, 1998.
15. Belmonti V, Cioni G, Berthoz A: Anticipatory control and spatial cognition in locomotion and navigation through typical development and in cerebral palsy. Dev Med Child Neurol 58 Suppl 4:22–27, 2016.
16. Dale N, Sakkalou E, O'Reilly M, et al.: Functional vision and cognition in infants with congenital disorders of the peripheral visual system. Dev Med Child Neurol 59:725–731, 2017.
17. Sonksen PM, Petrie A, Drew KJ: Promotion of visual development of severely visually impaired babies: evaluation of a developmentally based programme. Dev Med Child Neurol 33:320–335, 1991.
18. Atkinson J, Braddick O: Visual and visuocognitive development in children born very prematurely. Prog Brain Res 164:123–149, 2007.
19. Bushneil J, Sai F, Mullin J: Neonatal recognition of the mother's face. Br J Dev Psychol 7:3–15, 1989.
20. Yoon JM, Johnson SC: Biological motion displays elicit social behavior in 12-month-olds. Child Dev 80:1069–1075, 2009.
21. Braddick O, Atkinson J, Wattam-Bell J: Normal and anomalous development of visual

ACCEPTED MANUSCRIPT

motion processing: Motion coherence and “dorsal-stream vulnerability.” Neuropsychologia 41:1769–1784, 2003.

22. Dutton GN: Cognitive vision, its disorders and differential diagnosis in adults and children: knowing where and what things are. Eye (Lond) 17:289–304, 2003.
23. Aslin RN, Smith LB: Perceptual development. Annu Rev Psychol 39:435–473, 1988.
24. De Renzi E: Disorders of visual recognition. Semin Neurol 20:479–485, 2000.
25. Cioni G, Inguaggiato E, Sgandurra G: Early intervention in neurodevelopmental disorders: underlying neural mechanisms. Dev Med Child Neurol 58 Suppl 4:61–66, 2016.
26. Bova SM, Giovenzana A, Signorini S, et al.: Recovery of visual functions after early acquired occipital damage. Dev Med Child Neurol 50:311–315, 2008.
27. Ruberto G, Salati R, Milano G, et al.: Changes in the optic disc excavation of children affected by cerebral visual impairment: a tomographic analysis. Invest Ophthalmol Vis Sci 47:484–488, 2006.
28. Good W V, Jan JE, DeSa L, et al.: Cortical visual impairment in children. Surv Ophthalmol 38:351–364, 1994.
29. Saidkasimova S, Bennett DM, Butler S, et al.: Cognitive visual impairment with good visual acuity in children with posterior periventricular white matter injury: a series of 7 cases. J aapos 11:426–430, 2007.
30. Fazzi E, Bova S, Giovenzana A, et al.: Cognitive visual dysfunctions in preterm children with periventricular leukomalacia. Dev Med Child Neurol 51:974–981, 2009.
31. Ortibus E, Verhoeven J, Sunaert S, et al.: Integrity of the inferior longitudinal fasciculus and impaired object recognition in children: A diffusion tensor imaging study. Dev Med Child Neurol 54, 2012.
32. Bauer CM, Heidary G, Koo BB, et al.: Abnormal white matter tractography of visual pathways detected by high-angular-resolution diffusion imaging (HARDI) corresponds to visual dysfunction in cortical/cerebral visual impairment. J aapos 18:398–401, 2014.
33. Galli J, Ambrosi C, Micheletti S, et al.: White matter changes associated with cognitive

ACCEPTED MANUSCRIPT

visual dysfunctions in children with cerebral palsy: A diffusion tensor imaging study. J Neurosci Res, 2018.

34. Dutton GN: “Dorsal stream dysfunction” and “dorsal stream dysfunction plus”: a potential classification for perceptual visual impairment in the context of cerebral visual impairment? Dev Med Child Neurol 51:170–172, 2009.

35. Laporta-Hoyos O, Fiori S, Pannek K, et al.: Brain lesion scores obtained using a simple semi-quantitative scale from MR imaging are associated with motor function, communication and cognition in dyskinetic cerebral palsy. NeuroImage Clin 19:892–900, 2018.

36. Franki I, Fehrenbach A, Peedima M-L, et al.: Quantitative brain lesion scoring on neuroimaging in children with and without a diagnosis of cerebral visual impairment. Dev Med Child Neurol 59:20, 2017.

37. Goodale MA, Milner AD: Separate visual pathways for perception and action. Trends Neurosci 15:20–25, 1992.

38. Dutton GN, Jacobson LK: Cerebral visual impairment in children. Semin Neonatol 6:477–485, 2001.

39. Jakobson LS, Taylor NM: Differential vulnerability of cerebral visual functions in children born very prematurely. Acta Paediatr 98:239–241, 2009.

40. Gougoux F, Belin P, Voss P, et al.: Voice perception in blind persons: a functional magnetic resonance imaging study. Neuropsychologia 47:2967–2974, 2009.

41. Sargent J, Salt A, Dale N: Children with severe brain damage: Functional assessment for diagnosis and intervention. In: Visual Impairment in Children due to Damage to the Brain. 2010. .

42. Ortibus E, Laenen A, Verhoeven J, et al.: Screening for cerebral visual impairment: Value of a CVI questionnaire. Neuropediatrics 42, 2011.

43. Salavati M, Waninge A, Rameckers EA, et al.: Development and face validity of a cerebral visual impairment motor questionnaire for children with cerebral palsy. Child Care Heal Dev 43:37–47, 2017.

ACCEPTED MANUSCRIPT

44. Ferziger NB, Nemet P, Brezner A, et al.: Visual assessment in children with cerebral palsy: implementation of a functional questionnaire. Dev Med Child Neurol 53:422–428, 2011.

45. Macintyre-Beon C, Young D, Calvert J, et al.: Reliability of a question inventory for structured history taking in children with cerebral visual impairment. In: Eye (Lond). England; 2012. p. 1393. .

46. Philip SS, Tsherlinga S, Thomas MM, et al.: A Validation of an Examination Protocol for Cerebral Visual Impairment Among Children in a Clinical Population in India. J Clin Diagn Res 10:Nc01-4, 2016.

47. Garcia-Ormaechea I, Gonzalez I, Dupla M, et al.: Validation of the Preverbal Visual Assessment (PreViAs) questionnaire. Early Hum Dev 90:635–638, 2014.

48. Reynell J, Zinkin PM: New procedures for developmental assessment of young children with severe visual handicaps. Child Care Health Dev 1:61–69, 1975.

49. Good W V: Development of a quantitative method to measure vision in children with chronic cortical visual impairment. Trans Am Ophthalmol Soc 99:253–269, 2001.

50. Fazzi E, Signorini SG, Bova SM, et al.: Spectrum of visual disorders in children with cerebral visual impairment. J Child Neurol 22:294–301, 2007.

51. Fazzi E, Signorini SG, R LAP, et al.: Neuro-ophthalmological disorders in cerebral palsy: ophthalmological, oculomotor, and visual aspects. Dev Med Child Neurol 54:730–736, 2012.

52. Baker-Nobles L, Rutherford A: Understanding cortical visual impairment in children. Am J Occup Ther 49:899–903, 1995.

53. Jakobson LS, Frisk V, Downie AL: Motion-defined form processing in extremely premature children. Neuropsychologia 44:1777–1786, 2006.

54. Ricci D, Cesarini L, Groppo M, et al.: Early assessment of visual function in full term newborns. Early Hum Dev 84:107–113, 2008.

55. Ricci D, Cesarini L, Romeo DM, et al.: Visual function at 35 and 40 weeks' postmenstrual age in low-risk preterm infants. Pediatrics 122:e1193-8, 2008.

56. Leonhardt M, Forns M, Calderon C, et al.: Visual performance in preterm infants with brain injuries compared with low-risk preterm infants. Early Hum Dev 88:669–675, 2012.

57. Romeo DM, Ricci D, Serrao F, et al.: Visual function assessment in late-preterm newborns. Early Hum Dev 88:301–305, 2012.

58. Rossi A, Gnesi M, Montomoli C, et al.: Neonatal Assessment Visual European Grid (NAVEG): Unveiling neurological risk. Infant Behav Dev 49:21–30, 2017.

59. Teller DY, McDonald MA, Preston K, et al.: Assessment of visual acuity in infants and children: the acuity card procedure. Dev Med Child Neurol 28:779–789, 1986.

60. Salt AT, Wade AM, Proffitt R, et al.: The Sonksen logMAR Test of Visual Acuity: I. Testability and reliability. J aapos 11:589–596, 2007.

61. Sonksen PM: The assessment of vision in the preschool child. Arch Dis Child 68:513–516, 1993.

62. Koenraads Y, Braun KP, van der Linden DC, et al.: Perimetry in young and neurologically impaired children: the Behavioral Visual Field (BEFIE) Screening Test revisited. JAMA Ophthalmol 133:319–325, 2015.

63. Ushaw G, C S, J H, et al.: Analysis of Soft Data for Mass Provision of Stereoacuity Testing Through a Serious Game for Health. Int Conf Digit Heal (DH '17) 6:216–220, 2017.

64. Frisby J: The Frisby Stereotest: An introduction and a review. J Chem Inf Model 53:1689–1699, 2013.

65. Cavezian C, Vilayphonh M, de Agostini M, et al.: Assessment of visuo-attentional abilities in young children with or without visual disorder: toward a systematic screening in the general population. Res Dev Disabil 31:1102–1108, 2010.

66. Manly T, Anderson V, Nimmo-Smith I, et al.: The differential assessment of children's attention: the Test of Everyday Attention for Children (TEA-Ch), normative sample and ADHD performance. J Child Psychol Psychiatry 42:1065–1081, 2001.

67. Korkman M, Kirk U, Kemp SL: NEPSY II, 2nd ed. , 2007.

ACCEPTED MANUSCRIPT

68. Martin N.: Test of visual perceptual skills. In: Novato C, editor. Academic Therapy Publications. 2006. .

69. Brown T: Validity and Reliability of the Developmental Test of Visual Perception - Third Edition (DTVP-3). Occup Ther Heal Care 30:272–288, 2016.

70. Vancleef K, Acke E, Torfs K, et al.: Reliability and validity of the Leuven Perceptual Organization Screening Test (L-POST) The Leuven Perceptual Organization Screening Test (L-POST), an online test to assess mid-level visual perception. J Neuropsychol 9:271–298, 2015.

71. Torfs K, Vancleef K, Lafosse C, et al.: The Leuven Perceptual Organization Screening Test (L-POST), an online test to assess mid-level visual perception. Behav Res Methods 46:472–487, 2014.

72. Stiers P, Vanderkelen R, Vanneste G, et al.: Visual-perceptual impairment in a random sample of children with cerebral palsy. Dev Med Child Neurol 44:370–382, 2002.

73. Ortibus E, Peirens G, Schoolmeesters B, et al.: G(anspoel).CVI.Tod(dler)s: a novel diagnostic tool in the assessment of cerebral visual impairment in the young child. In: European Academy for Childhood Disability. Stockholm, Sweden: Mac Keith Press; 2016. p. 49. .

74. Ortibus E, Vancleef K, Baümer S, et al.: CVIT 3-6, a screening test for cerebral visual impairment in young children. In: European Society for Low Vision Research and Rehabilitation. Oxford, UK; 2015. .

75. Hard AL, Niklasson A, Svensson E, et al.: Visual function in school-aged children born before 29 weeks of gestation: a population-based study. Dev Med Child Neurol 42:100–105, 2000.

76. Braddick O, Atkinson J: Development of brain mechanisms for visual global processing and object segmentation. Prog Brain Res 164:151–168, 2007.

77. Verhoef BE, Vogels R, Janssen P: Binocular depth processing in the ventral visual pathway. Philos Trans R Soc L B Biol Sci 371, 2016.

78. Atkinson J, Braddick O: From genes to brain development to phenotypic behavior:

ACCEPTED MANUSCRIPT

"dorsal-stream vulnerability" in relation to spatial cognition, attention, and planning of actions in Williams syndrome (WS) and other developmental disorders. Prog Brain Res 189:261–283, 2011.

79. Braddick O, Atkinson J: Visual control of manual actions: Brain mechanisms in typical development and developmental disorders. Dev Med Child Neurol 55:13–18, 2013.
80. Belmonti V, Berthoz A, Cioni G, et al.: Navigation strategies as revealed by error patterns on the Magic Carpet test in children with cerebral palsy. Front Psychol 6:880, 2015.
81. van Hartingsveldt MJ, Cup EH, Oostendorp RA: Reliability and validity of the fine motor scale of the Peabody Developmental Motor Scales-2. Occup Ther Int 12:1–13, 2005.
82. Croce R V, Horvat M, McCarthy E: Reliability and concurrent validity of the movement assessment battery for children. Percept Mot Ski 93:275–280, 2001.
83. Beery K: The Beery-Buktenica Developmental Test of Visual-Motor Integration: VMI with Supplemental Developmental Tests of Visual Perception and Motor Coordination: Administration, Scoring and Teaching Manual. , 1997.
84. Borchers S, Muller L, Synofzik M, et al.: Guidelines and quality measures for the diagnosis of optic ataxia. Front Hum Neurosci 7:324, 2013.
85. Williams C, Gilchrist ID, Fraser S, et al.: Normative data for three tests of visuocognitive function in primary school children: cross-sectional study. Br J Ophthalmol 99:752–756, 2015.
86. Belmonti V, Fiori S, Guzzetta A, et al.: Cognitive strategies for locomotor navigation in normal development and cerebral palsy. Dev Med Child Neurol 57 Suppl 2:31–36, 2015.
87. Dutton GN: Vision and the brain. Understanding Cerebral Visual Impairment in Children. AFB Press; 2015.
88. Colenbrander A: Assessment of functional vision and its rehabilitation. Acta Ophthalmol 88:163–173, 2010.

ACCEPTED MANUSCRIPT

89. Grinter EJ, Maybery MT, Badcock DR: Vision in developmental disorders: is there a dorsal stream deficit? Brain Res Bull 82:147–160, 2010.

90. Van der Hallen R, Evers K, Brewaeys K, et al.: Global processing takes time: A meta-analysis on local-global visual processing in ASD. Psychol Bull 141:549–573, 2015.

91. Noten M, Wilson P, Ruddock S, et al.: Mild impairments of motor imagery skills in children with DCD. Res Dev Disabil 35:1152–1159, 2014.

92. Chokron S, Dutton GN: Impact of cerebral visual impairments on motor skills: Implications for developmental coordination disorders. Front Psychol 7:1–15, 2016.

93. Pawletko T, Chokron S, Dutton G: Considerations in the Behavioral Diagnosis of CVI: Issues, Cautions and Potential Outcomes. In: Lueck AH, Dutton GN, editors. Vision and the Brain. NY, USA: AFB Press; . .

94. Atkinson J: The Davida Teller Award Lecture, 2016: Visual Brain Development: A review of “Dorsal Stream Vulnerability”-motion, mathematics, amblyopia, actions, and attention. J Vis 17:26, 2017.

95. Philip SS, Dutton GN: Identifying and characterising cerebral visual impairment in children: A review. Clin Exp Optom 97:196–208, 2014.

96. Vervloed MP, Janssen N, Knoors H: Visual rehabilitation of children with visual impairments. J Dev Behav Pediatr 27:493–506, 2006.

97. Berardi N, Sale A, Maffei L: Brain structural and functional development: genetics and experience. Dev Med Child Neurol 57 Suppl 2:4–9, 2015.

98. Chorna OD, Guzzetta A, Maitre NL: Vision Assessments and Interventions for Infants 0-2 Years at High Risk for Cerebral Palsy: A Systematic Review. Pediatr Neurol 76:3–13, 2017.

99. Chen YN, Lin CK, Wei TS, et al.: The effectiveness of multimedia visual perceptual training groups for the preschool children with developmental delay. Res Dev Disabil 34:4447–4454, 2013.

100. Sagi D: Perceptual learning in Vision Research. Vis Res 51:1552–1566, 2011.

101. Lawton T, Shelley-Tremblay J: Training on Movement Figure-Ground Discrimination

ACCEPTED MANUSCRIPT

Remediates Low-Level Visual Timing Deficits in the Dorsal Stream, Improving High-Level Cognitive Functioning, Including Attention, Reading Fluency, and Working Memory. Front Hum Neurosci 11:236, 2017.

102. Green CS, Bavelier D: Effect of action video games on the spatial distribution of visuospatial attention. J Exp Psychol Hum Percept Perform 32:1465–1478, 2006.

103. Green CS, Li R, Bavelier D: Perceptual learning during action video game playing. Top Cogn Sci 2:202–216, 2010.

104. Smits-Engelsman B, Vincon S, Blank R, et al.: Evaluating the evidence for motor-based interventions in developmental coordination disorder: A systematic review and meta-analysis. Res Dev Disabil 74:72–102, 2018.

105. Downs J, Blackmore AM, Epstein A, et al.: The prevalence of mental health disorders and symptoms in children and adolescents with cerebral palsy: a systematic review and meta-analysis. Dev Med Child Neurol 60:30–38, 2018.

106. Mathewson KJ, Chow CH, Dobson KG, et al.: Mental health of extremely low birth weight survivors: A systematic review and meta-analysis. Psychol Bull 143:347–383, 2017.

107. Li L, Liu J: The effect of pediatric traumatic brain injury on behavioral outcomes: a systematic review. Dev Med Child Neurol 55:37–45, 2013.

ACCEPTED MANUSCRIPT